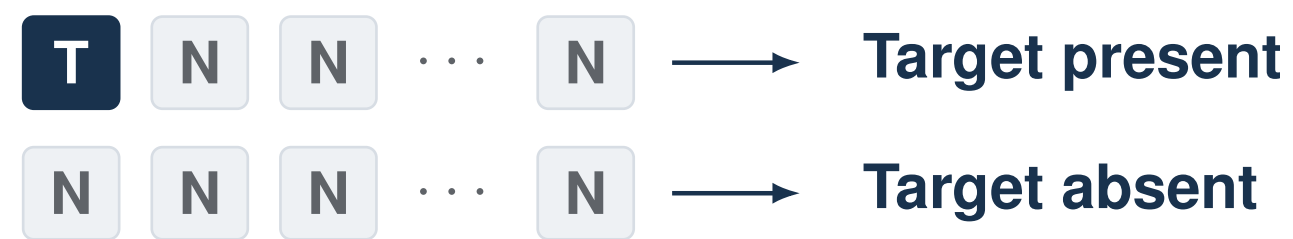


Motivation

Training on short sequences does not guarantee reliable performance on much longer inputs [1]. As sequence length grows, more non-target tokens compete for the same attention budget. We ask how attention must sharpen with length to remain concentrated on the target.

Task & Setup

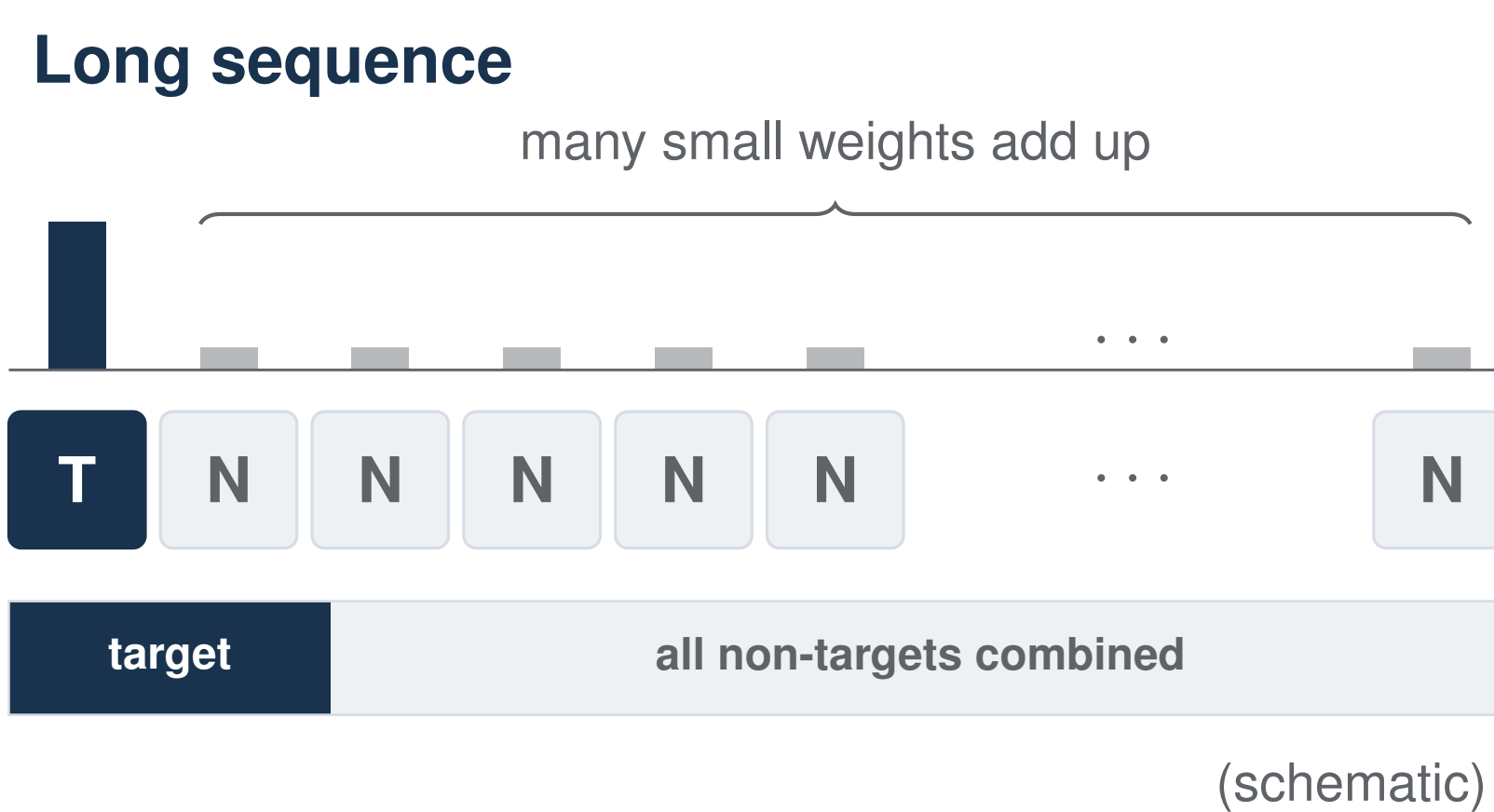
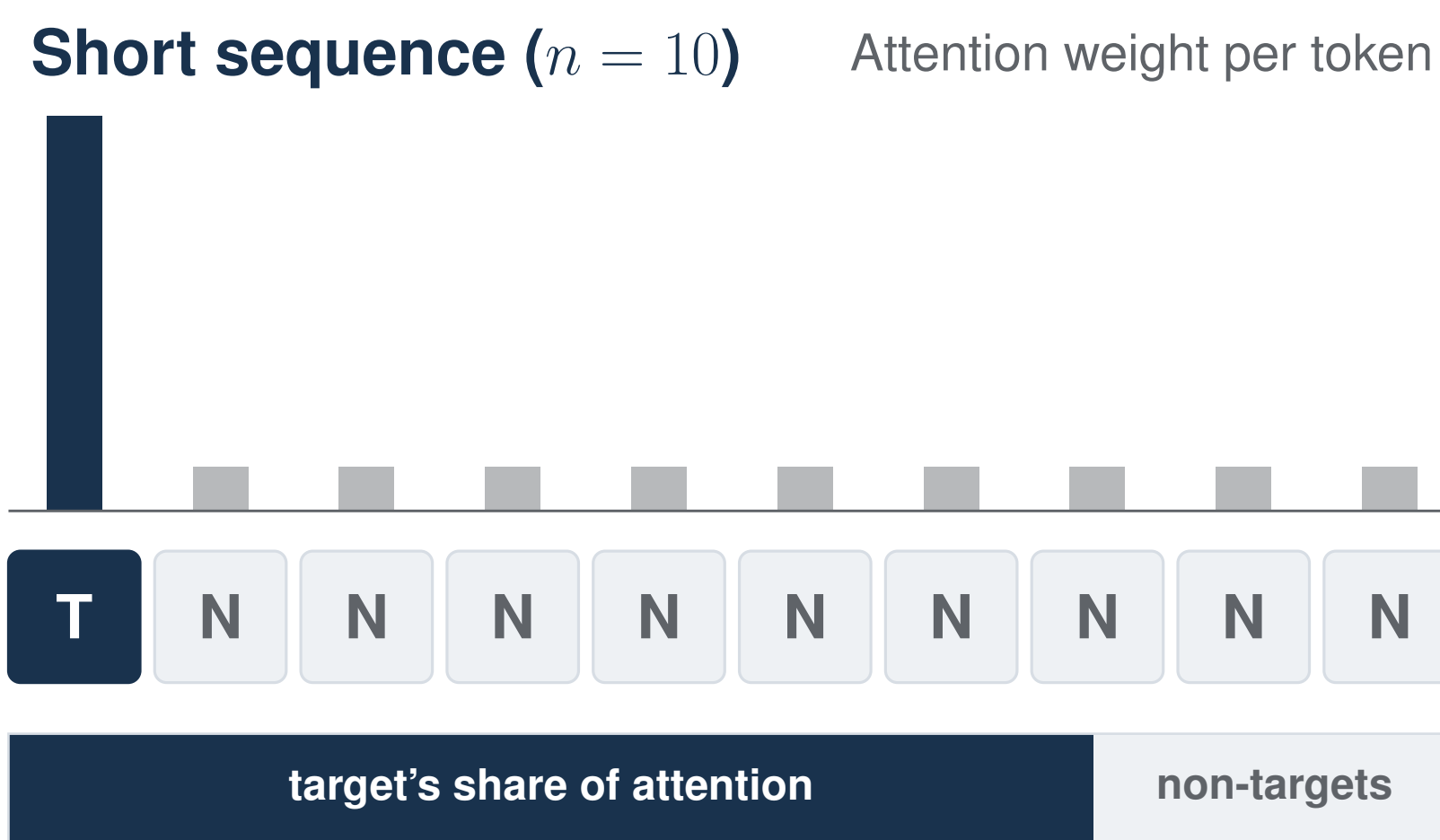
Detect whether a target token (**T**) is present. All other tokens are identical non-targets (**N**).



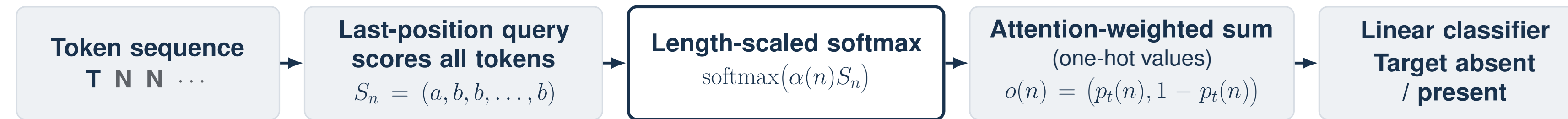
We use a reduced attention-based binary classifier with one readout query, fixed one-hot token values, and a linear output layer.

Why Attention Dilutes

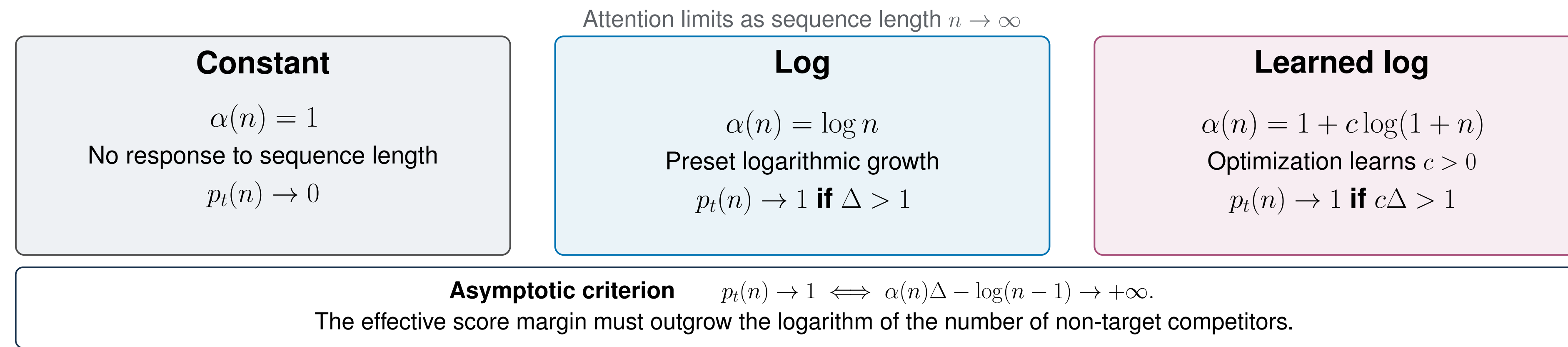
A fixed score advantage over each non-target cannot prevent the target's share of attention from shrinking as the sequence grows.



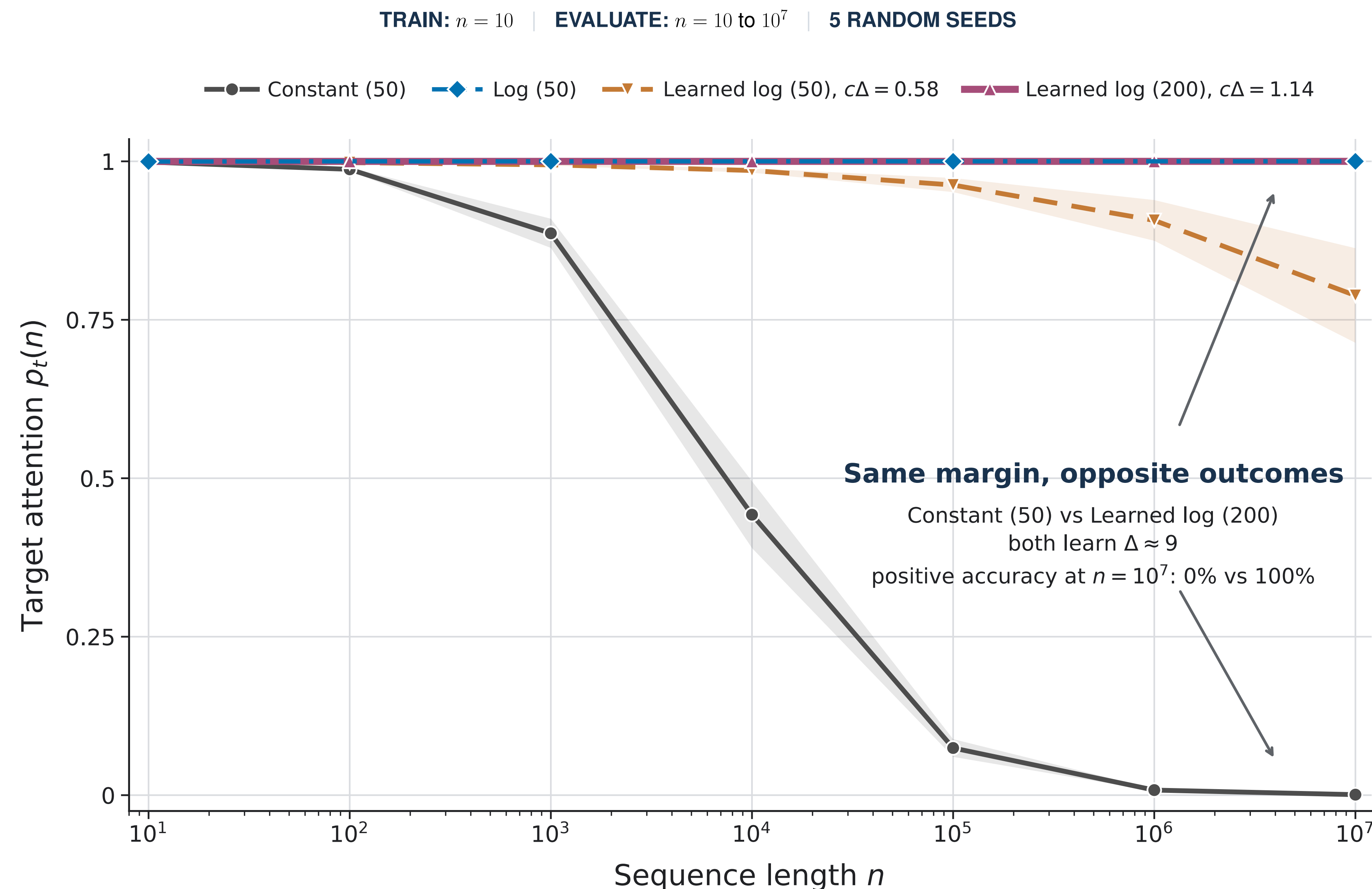
How Length-Aware Scaling Counters Dilution



Learned query and key vectors produce the pre-softmax scores a (target) and b (non-target) through scaled dot products. Their difference is the score margin $\Delta = a - b$. Multiplying every score by a positive factor $\alpha(n)$ preserves the score ranking and changes this margin to $\alpha(n)\Delta$.



Same Training-Length Accuracy, Different Long-Length Behavior



Means over five seeds; bands show ± 1 s.d.

Attention in Closed Form

Standard softmax [2] gives the target's attention share in a positive sequence of length n :

$$p_t(n) = \frac{e^{\alpha(n)a}}{e^{\alpha(n)a} + (n-1)e^{\alpha(n)b}}$$

Conclusion

- Even a large score advantage **cannot prevent attention dilution** under constant scaling.
- Sufficiently strong logarithmic scaling** makes the target receive nearly all attention as sequences grow.

Finite success can still hide later failure. Learned log (50) reaches 100% accuracy at ten million tokens, yet the theory predicts eventual failure.

Scope

The exact analysis applies to a reduced classifier with one target per positive sequence, identical non-targets, and fixed one-hot values.

Whether the same scaling conditions extend to full transformers remains open.

References & Acknowledgments

- [1] Press et al. (2022). *Train Short, Test Long: Attention with Linear Biases Enables Input Length Extrapolation*. ICLR.
- [2] Vaswani et al. (2017). *Attention Is All You Need*. NeurIPS.

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